Astronaut Game

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Applications and Databases

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1.Introduction

In this Astronaut Game, it was an interactive web-based game platform with user controls system which navigate through rocks obstacles and stars collection withing appropriate key control with leveling system and one special feature. This game involves login and logout for users, a comprehensive scoreboard features when game is over, player game manual for user friendly experience and then a dynamic leaderboard.

2. Features

2.1 User Authentication

Users are required to complete a form with their username, email address, phone number, home address, password, and password confirmation in order to register. If users forget to write something down, an alert message will appear. However, existing user cannot register which check through based on player's name.

Users can also Login with their given username and password from register . If the password or username is wrong the alert message will be shown.

All of the user's information from register are stored in the localstorage.

2.2 Gameplay Mechanics

- Controls : Player can control the astronaut using a specific set of keys for different actions which are as follows:
  - Double Jump : Press D for performing double jump to reach higher.
  - Jump : Press Space Bar to make astronaut jump.
  - Crawl : Press C for crawling which allow astronaut to evade low obstacle rocks.
  - Flight Mode : The astronaut game consists of collecting stars features, after collecting 80 stars, press F to activate flight mode which will last for 3 seconds and that will allow free movement to provide more score which pass over all of the obstacle rocks.

2.3 Scoring System

In this part, the information of users are kept in HTML local storage using JSON format. Once users complete the game, they will immediately be able to view their current scores in a popup message. Additionally, if they wish to view the scores of various players, they can check the scoreboard table in the navigation bar.

Top Score : Each player's high score is saved in the local storage and displayed after game is over which will displays top 5 players by score.

Latest Score : The latest score is saved in the localstorage and displayed after game is over.

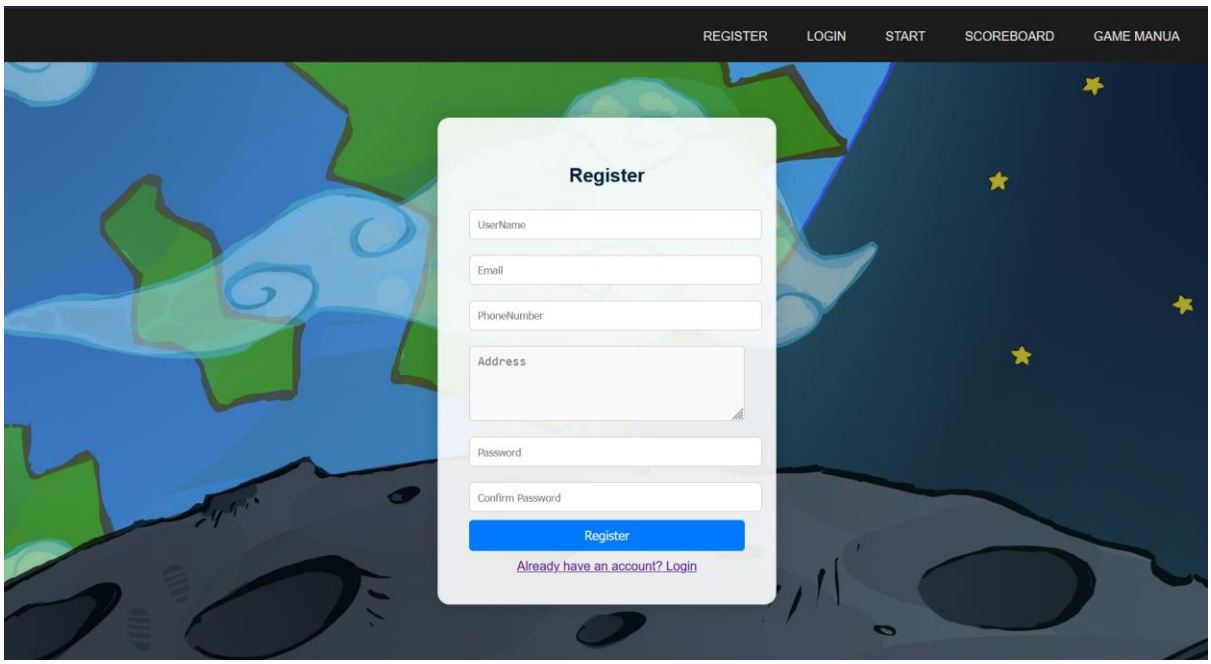
Scoreboard : The scoreboard will show top 10 players which will also accessible through navigation bar as well. If the user is log in, it will also show the current user's top score and latest score. If the user is log out, it will only show top 10 players with score.

2.4 Game Manual

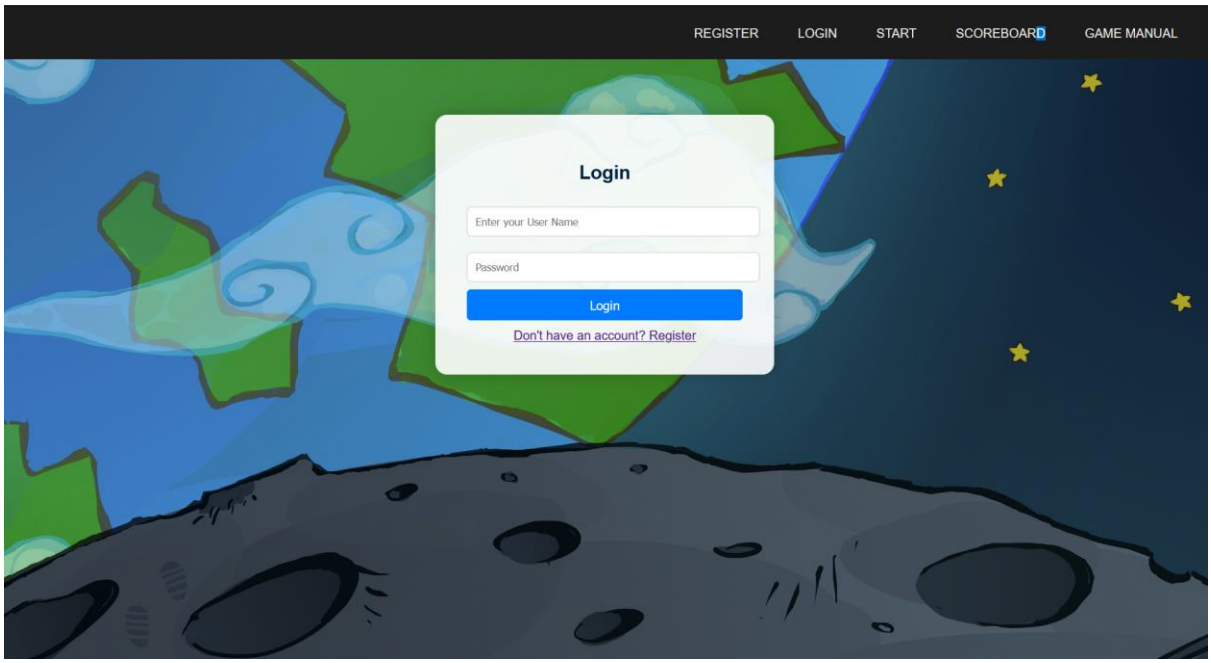
- It will provide user to know how to play game with the following instructions and key controls.
- To ensure players to learn about collecting stars and flight mode open which is the advanced tips.

3. User Interface

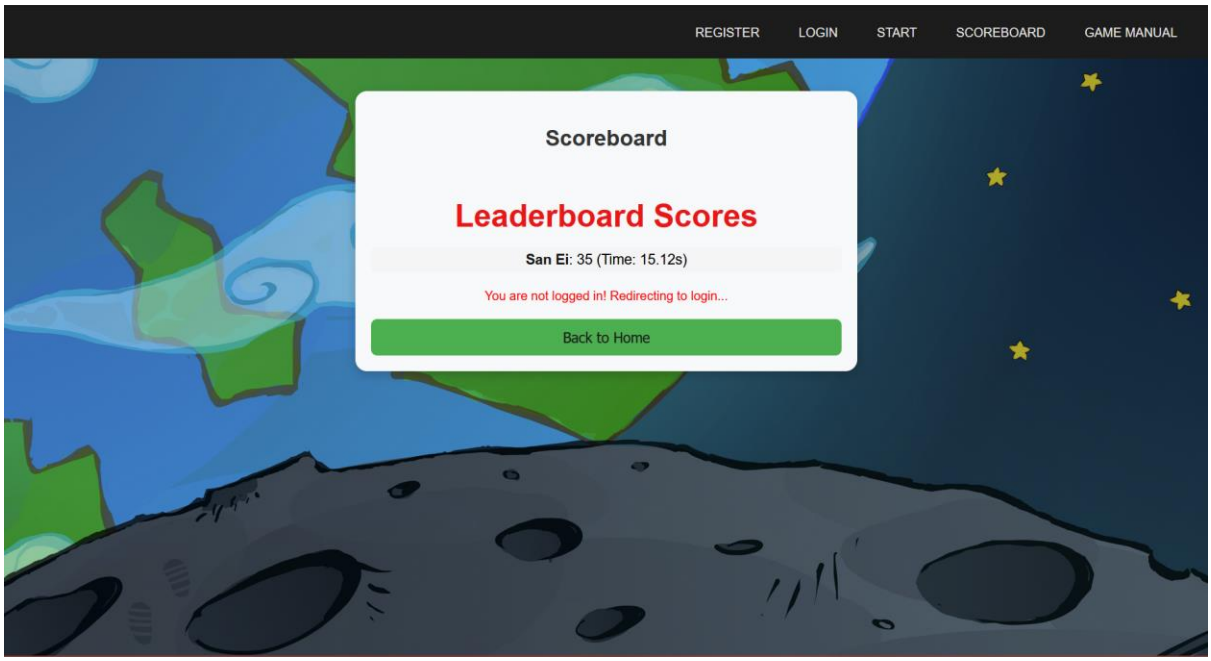
3.1 Register Page



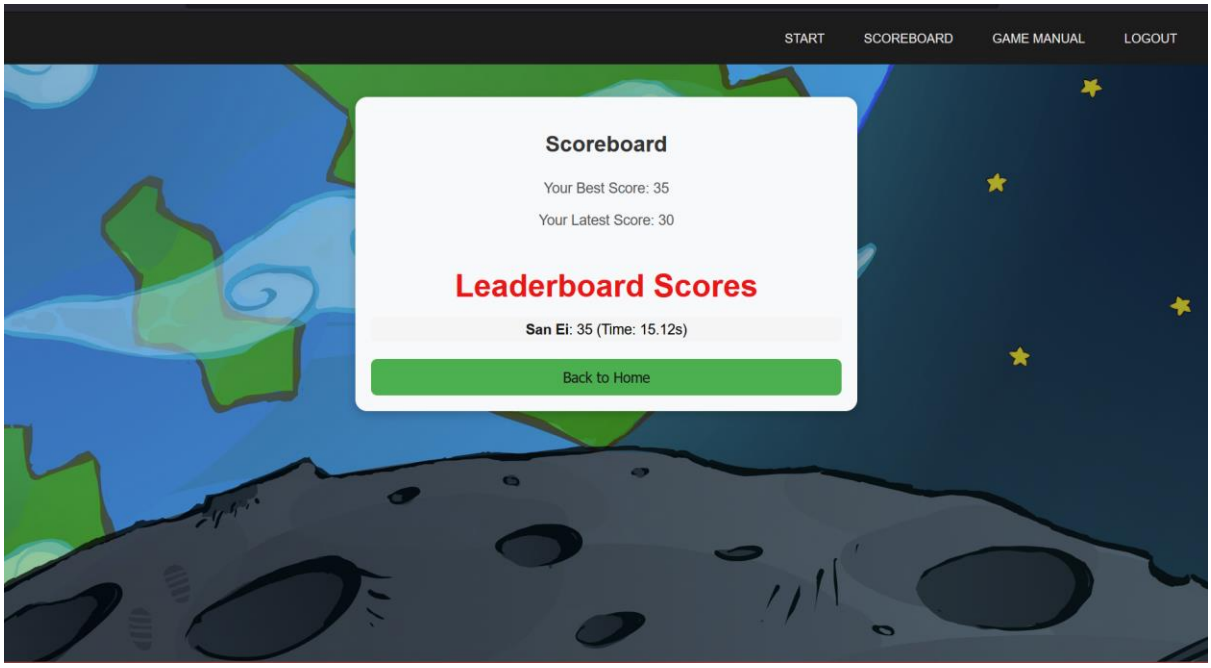
3.2 Login Page



3.3 Scoreboard Page

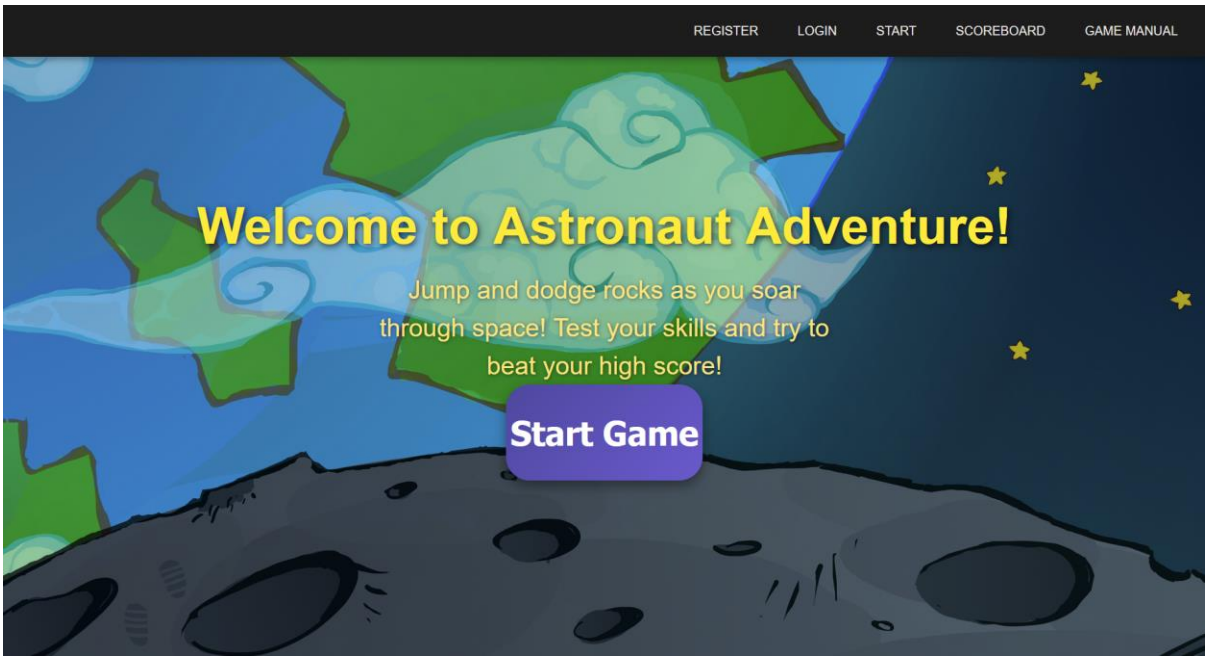


Scoreboard Page display when User is not logged in

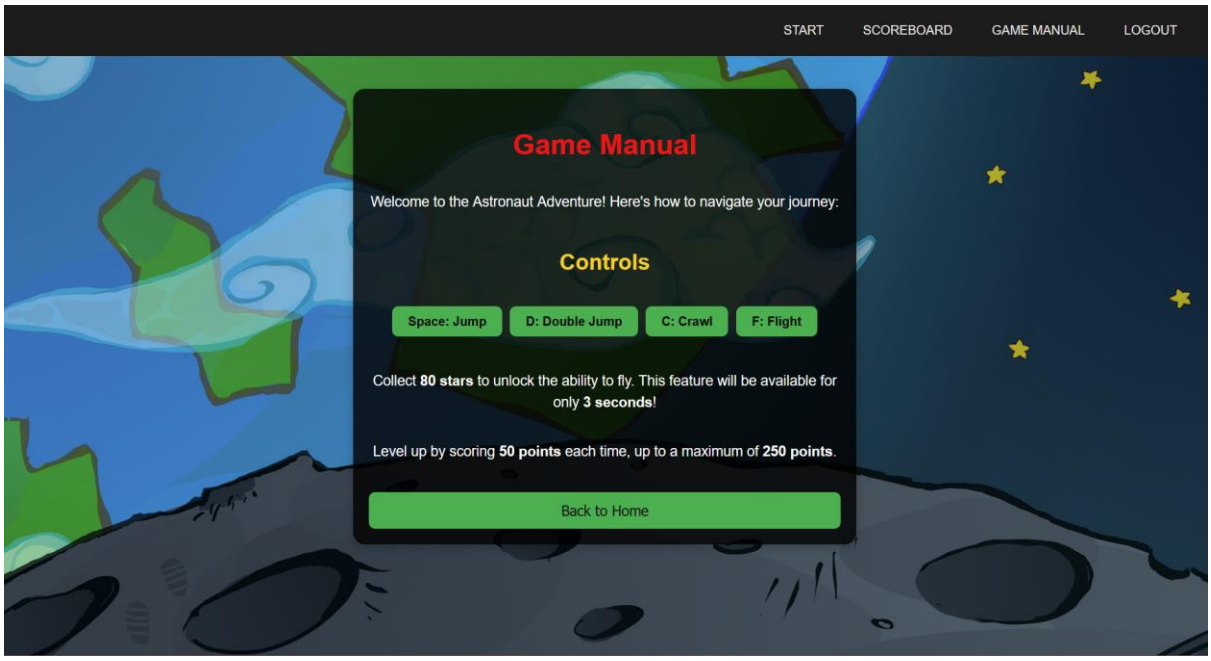


Scoreboard Display when user is logged in

3.4 Home Page



3.5 Game Manual Page

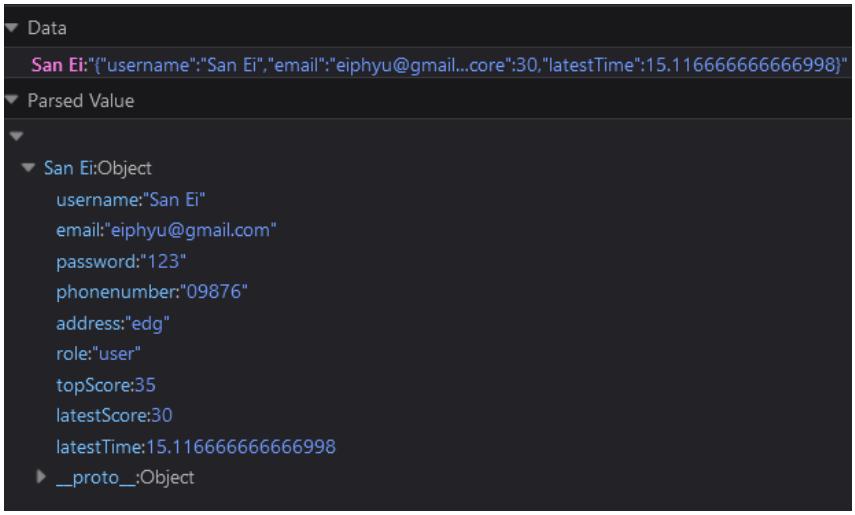


4. Authentication Flow

4.1 Login and User Registration

Login : when user is logged in, their data is retrieved and stored in local storage as follows:

- currentUser : It will store the active username.
- loggedIn : A boolean to indicate the login status
- logOut : It will clear the currentUser and loggedIn data from the local storage.
- users : It will hold users' specific data which include username, email address, phone number, address, password and confirm password and moreover, topScore, latestScore and latestTime will also store in the same Json format users local storage for more ease of storing data.



Example of storing information in the localStorage

4.2 Scoring and Leaderboard Management

- Score Calculation : Score and stars calculation will increase with real time based and also displayed when user is playing through the game which is crucial for user friendly.
- Leaderboard : It will update after each session of game play and display top 5 users when game is over.
- Scoreboard : The scoreboard will display top 10 players and current login user best score and latest score if they loggedin.

7. Error Handling

- LogOut redirection : If users are not logged in or after being logged out, it will redirect automatically to login page.
- Store Retrieval : Validate data correctly structured in local storage.
- Navbar update according to user Login Status : If user is logged in, no Register Page and Login Page at the navigation bar is not displayed which is hide with using user management.
- Alert user to login : If user is not logged in and start game, it will alert message user to login.

8. Challenges Encountered

8.1 LocalStorage Management

Challenge: Managing player data reliably in localStorage was more complex than expected.

Solution: We established a structured method to handle each user's profile in JSON format, using consistent keys (e.g., username_topScore, username_latestScore) to make access easier.

8.2 Handling User Authentication and Session Management

Challenge: Managing multiple user logins in the same browser session posed risks of data conflicts and inconsistencies.

Solution: We added a currentUser key in localStorage to identify the active user, along with a loggedIn key to monitor session status. These keys are updated with each login and logout, ensuring data consistency and secure access across different pages.

8.3 Leaderboard Sorting

Challenge: Updating scores in real-time while keeping an accurate leaderboard order for all users was difficult.

Solution: To resolve issues with ties, we introduced a latestTime property for each user and sorted the leaderboard first by score (in descending order) and then by latestTime (in ascending order) for accurate ranking.

8.4 Managing Multiple Controls

Challenge: The various controls (jump, double jump, crawl, and flight) led to conflicts when players switched actions quickly.

Solution: We implemented a state machine to organize control functions, assigning specific conditions to each action independently to avoid interference between controls.

8.5 Creating a Clear Interface for Score Updates and Game Over

Challenge: Presenting score updates and game-over screens in a user-friendly manner was challenging, as sudden changes or unclear information could confuse players.

Solution: We created a simple overlay for the Game Over screen that displays the latest score along with a clear option to view the leaderboard.

9. Conclusion

All in all, the web game offers a simple and easy-to-use experience with a navigation bar that lets users register, log in, start playing, check the scoreboard, read the game manual, and log out. The gameplay itself is designed around a simple yet challenging concept: players earn points by navigating through space and avoiding contact with floating stones. As players advance and level up, the game becomes more difficult, with increased speeds making it tougher to achieve high scores.

The game focuses on providing a positive experience, allowing players to easily access all key features from the navigation bar, even while playing. For managing data, user information like username, email, phone number, address, and password is securely stored in local storage using JSON format, which allows for easy validation and retrieval of data. Overall, the game combines engaging gameplay with a user-friendly design and efficient data handling to deliver an enjoyable experience.